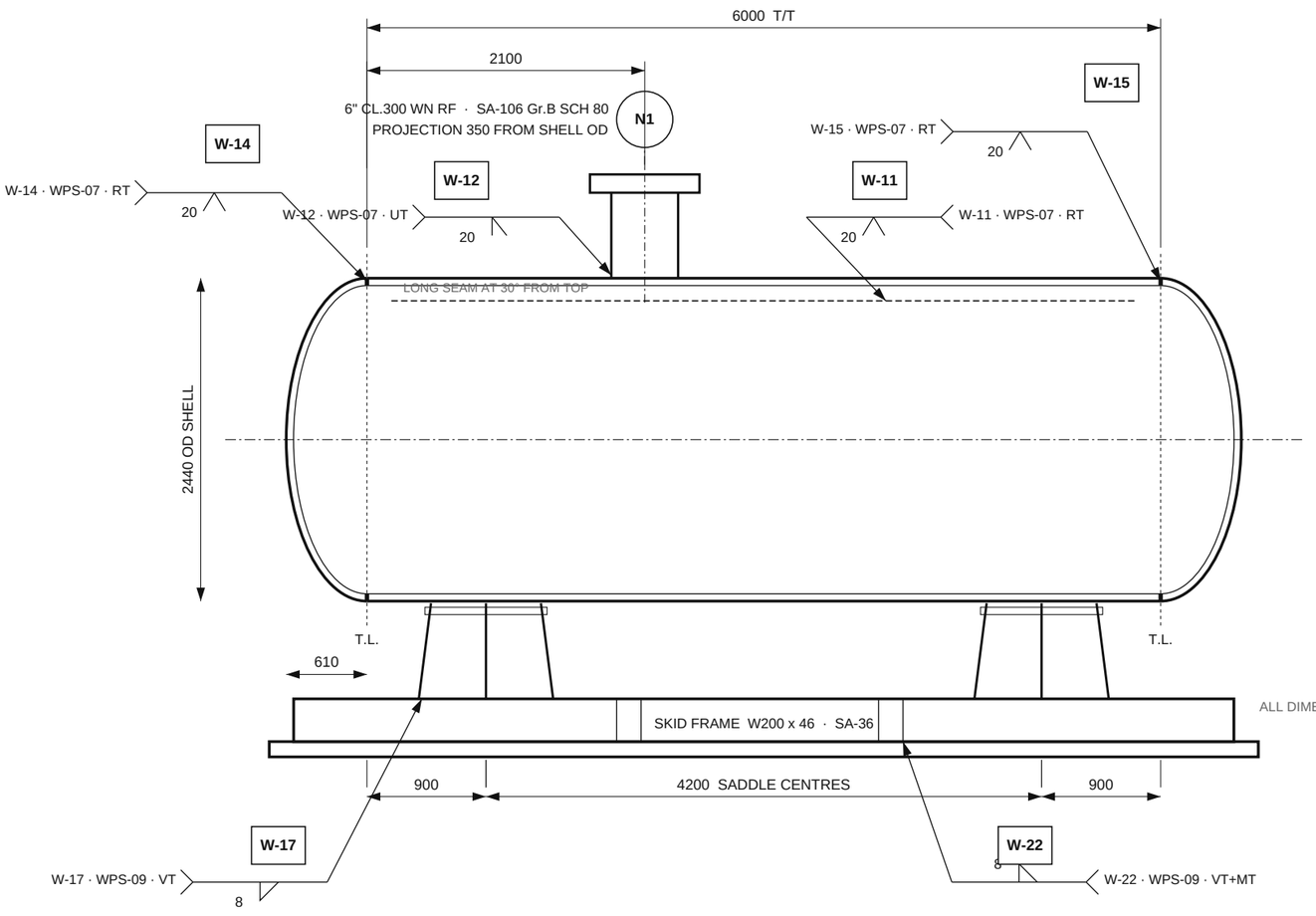
SYNTHETIC DOCUMENT - generated for software testing - not a real drawing

ELEVATION · NOT TO SCALE



ALL DIMENSIONS IN mm FROM LEFT T.L.

WELD MAP / WELD TABLE

Joint	Description	Cat.	Type	t (mm)	OD (mm)	Pos.	WPS	Welder	NDE	PWHT
W-11	Shell long seam	A	Groove	20	2440	1G	WPS-07	118	RT 100%	Yes
W-12	Nozzle N1 (6in) to shell	D	Groove	20	168.3	6G	WPS-07	118	UT 100%	Yes
W-14	Shell to head, left (circ)	B	Groove	20	2440	1G	WPS-07	118	RT 100%	Yes
W-15	Shell to head, right (circ)	B	Groove	20	2440	1G	WPS-07	118	RT 100%	Yes
W-17	Skid beam to base plate	-	Fillet	12	n/a	2F	WPS-09	204	VT 100%	No
W-22	Skid cross beam, overhead	-	Fillet	12	n/a	4F	WPS-09	204	VT 100% + MT 10%	No

MATERIAL LIST

Item	Part	Material	t (mm)	Heat no.
1	Shell, 1 course	SA-516 Gr.70 N	20	4A7721
2	Heads, 2:1 SE (2)	SA-516 Gr.70 N	20	4A7721
3	Nozzle neck N1, 6in SCH 80	SA-106 Gr.B	10.97	5B2210
4	Skid beams and base plates	SA-36	12	3C0954

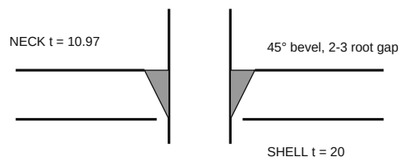
DESIGN DATA

Code	ASME BPVC VIII-1, 2023 edition
Shell	2400 ID / 2440 OD x 20 mm (no process piping on this drawing)
Design pressure	1.6 MPa (16 bar)
Design temperature	210 °C
MDMT	-20 °C at 1.6 MPa
Corrosion allowance	3 mm
Joint efficiency	E = 1.0 (full RT, Cat. A and B)
Hydrotest	UG-99(b): 1.3 x MAWP = 2.08 MPa minimum, hold 30 min
PWHT	Required by service (caustic, client spec NR-210): 620 ± 15 °C, 1 h per 25 mm

NOTES

1. Construction per ASME BPVC VIII-1, 2023 edition. Welding per ASME IX; NDE per ASME V.
2. Category A and B butt welds: full radiography per UW-11(a). Nozzle N1 weld: UT 100 %.
3. PWHT of the vessel (W-11, W-12, W-14, W-15) after all pressure welding; no welding after PWHT.
4. Skid frame (W-17, W-22) is a non-pressure structure; vessel saddles are bolted to it. No PWHT.
5. Weld IDs in boxes and weld-symbol tails refer to the weld table. Production positions as listed.

DETAIL A · N1 SET-IN, FULL PENETRATION (W-12)



PRESSURE VESSEL V-110 ON SKID · SHOP DRAWING / WELD MAP	
Drawing no. **HBK-PV-110**	Revision **B**
Client **Northfield Refining (fictional)**	Scale **NTS**
Drawn **M. Tanaka (fictional) 2026-09-18**	Checked **R. Lindqvist (fictional) 2026-09-19**